



Patient Autonomy over Electronic Medical Records after Death: Reconstructing the Post-Mortem Privacy Legal Framework through Blockchain Technology

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Abstract

The advancement of digital healthcare has transformed Electronic Medical Records (EMRs) into valuable health data containing highly sensitive personal information. However, Indonesian law has not provided a clear mechanism regarding access to EMRs after a patient's death, particularly concerning the authority to determine access and the continued recognition of patients' wishes expressed during their lifetime. This study aims to analyze the legal framework governing post-mortem access to EMRs, examine the relevance of patient autonomy and post-mortem privacy, and formulate a blockchain-based legal reconstruction model for EMR governance. This research employs normative legal research using a qualitative approach through statutory, conceptual, and comparative approaches. The findings indicate that existing regulations in Indonesia primarily emphasize confidentiality but have not sufficiently accommodated patient autonomy in determining post-mortem access to health data. Therefore, a legal reconstruction is required through the recognition of electronic advance directives as a mechanism for expressing patients' preferences regarding access to their EMRs after death. Blockchain technology is proposed as a supporting instrument to ensure data integrity, authentication, auditability, and accountability through secure consent management mechanisms. This model aims to balance patient rights, legal certainty for healthcare providers, and protection of post-mortem privacy within Indonesia's digital healthcare system.

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I. INTRODUCTION

The death of an individual does not eliminate the existence of Electronic Medical Records (EMRs) containing highly sensitive health data. However, Indonesian law has not provided a clear legal framework regarding who has the authority to determine access to such data after a patient's death or the extent to which the patient's wishes expressed during their lifetime should continue to be respected. In the era of digital health transformation, Electronic Medical Records are no longer regarded merely as administrative documents but also as valuable health data that support medical services, healthcare financing, research, and the development of artificial intelligence (AI)-based healthcare technologies. (Mukharram, Nurita, & Paramarta, 2024). As the use of health data continues to expand, so does the need for legal protection that ensures the confidentiality and security of patient information. However,

when a patient dies, a fundamental legal question arises that has yet to be clearly addressed under the Indonesian legal system: who has the authority to determine access to the deceased patient's Electronic Medical Records, and whether the patient's wishes expressed during their lifetime should continue to be respected after death.

As a country actively promoting the digitalization of healthcare services through the implementation of Electronic Medical Records (EMRs) and the integration of the SATUSEHAT platform, Indonesia has enacted various legal instruments to protect patients' health data, including Law Number 17 of 2023 on Health, Law Number 27 of 2022 on Personal Data Protection, and Minister of Health Regulation Number 24 of 2022 on Medical Records. Article 32(1) of Minister of Health Regulation Number 24 of 2022 expressly provides that the contents of medical records must remain confidential and be protected by all parties involved in the provision of healthcare services, even after the patient's death. (Tiorentap, 2020). This provision demonstrates that Indonesian law protects the confidentiality of Electronic Medical Records (EMRs) even after a patient's death. However, protecting confidentiality does not automatically equate to protecting patient autonomy. To date, there is no legal mechanism that allows individuals to determine during their lifetime who may or may not access their Electronic Medical Records after their death. Consequently, respect for patients' wishes has not yet received adequate normative recognition under Indonesian law.

The absence of such a legal framework gives rise to various legal issues that may lead to conflicts of interest. In practice, heirs may require access to medical records for the resolution of legal disputes, insurance claims, or other legitimate legal purposes. At the same time, hospitals are under a legal obligation to maintain the confidentiality of medical records, while the deceased patient may have expressed specific wishes regarding who should or should not have access to their health information after death. Indonesian positive law has yet to provide clear guidance on the legal status of such wishes, the scope of heirs' authority, or the procedures for granting access to health data following a patient's death. Consequently, the obligation to protect the confidentiality of medical records may conflict with respect for the patient's wishes, the interests of family members, and the needs of healthcare services and law enforcement.

This issue should not be regarded merely as a matter of healthcare administration but also as one that concerns the fundamental principle of patient autonomy. In health law and bioethics, patient autonomy refers to the right of every individual to make decisions concerning their own body, medical treatment, and information relating to their health condition. Numerous studies have shown that respecting patient autonomy improves the quality of healthcare services and strengthens patient-centered decision-making (Seh et al., 2020). However, most of these studies limit the application of patient autonomy to the period during which an individual is still alive. The rapid development of digital technology, however, has given rise to a new legal issue: whether the right to determine the use of

personal health information should continue to be respected after a patient's death. This conceptual gap has received little attention in Indonesian health law scholarship..

Within contemporary international legal discourse, this issue has increasingly been addressed through the concept of post-mortem privacy, which recognizes that death does not automatically extinguish all legal interests relating to an individual's personal information (Harbinja, Morse, & Edwards, 2026). The debate surrounding digital legacy indicates that individuals generally have preferences regarding the management of their digital assets after death. However, there are still no legal or technological mechanisms capable of ensuring that such wishes are effectively implemented. Electronic Medical Records (EMRs) form part of an individual's digital identity and contain some of the most sensitive personal information. Therefore, the issue of post-mortem access to such data extends beyond medical confidentiality and also concerns respect for human dignity, the right to informational self-determination, and the protection of privacy in the digital environment (Holt, Smeddinck, Nicholson, Vlachokyriakos, & Durrant, 2026).

A comparison with Malaysia demonstrates that the legal protection of the personal data of deceased individuals remains limited. Under the Personal Data Protection Act 2010 and its subsequent regulatory developments, personal data protection is, in principle, afforded only to living individuals; consequently, data relating to deceased persons are not recognized as belonging to a data subject (Ghani, Shabri, Rasli, Razali, & Shuffri, 2020). On the other hand, Malaysia recognizes respect for an individual's wishes through the Human Tissues Act 1974, which governs the use of human tissue based on consent given during the individual's lifetime or the authority of designated persons after death (Ismail, 1996). This experience demonstrates that, although the concept of post-mortem privacy has not been explicitly recognized, legal efforts have been made to accommodate individuals' wishes in certain contexts. This comparison suggests that Indonesia likewise requires a clearer legal framework governing the relationship between patients' wishes, the rights of family members, and the obligations of healthcare providers (Khoo & Ballesté, 2024).

Research on Electronic Medical Records (EMRs) in Indonesia has generally focused on personal data protection, system security, and legal liability for data breaches. Meanwhile, studies on blockchain technology have primarily examined issues of data security and integrity, whereas research on patient autonomy has largely developed within the context of medical decision-making during a patient's lifetime (Komalasari & Mustafa, 2024). Consequently, there remains a significant research gap in integrating the principle of patient autonomy, the concept of post-mortem privacy, and the governance of post-mortem access to Electronic Medical Records within a coherent legal framework (Mandey, 2025a). This gap demonstrates that the central issue is no longer merely how to preserve the confidentiality of health data, but rather how the law should provide patients with the opportunity to determine the future use of their health information after death.

Based on the foregoing, this study argues that the current Indonesian legal framework has not yet fully accommodated patient autonomy with respect to post-mortem Electronic Medical Records (EMRs). Therefore, a reconstruction of the legal framework is required not only to preserve the principle of medical record confidentiality but also to establish a mechanism through which patients' wishes can be expressed, verified, and implemented in an accountable manner. In this context, blockchain technology is not positioned as the primary focus of this study but rather as a supporting instrument for recording digital consent, providing an immutable audit trail, and enhancing legal certainty in authorizing access to Electronic Medical Records (Santoso, Soraes, Gegen, & Astuti, 2024). This study aims to analyze the regulatory gaps governing post-mortem access to Electronic Medical Records (EMRs), examine the relevance of the concept of post-mortem privacy in reinforcing patient autonomy, and formulate a blockchain-based model for the reconstruction of the legal framework to advance privacy protection, legal certainty, and accountability within Indonesia's digital healthcare system.

II. RESEARCH METHODS

This study employs normative legal research using a qualitative approach and a library research method. The research adopts a statutory approach to examine Law Number 17 of 2023 on Health, Law Number 27 of 2022 on Personal Data Protection, Minister of Health Regulation Number 24 of 2022 on Medical Records, as well as relevant regulations in Malaysia. In addition, a conceptual approach is employed to analyze the concepts of patient autonomy, post-mortem privacy, and blockchain technology, while a comparative approach is used to compare the legal framework governing post-mortem access to Electronic Medical Records in Indonesia and Malaysia as the basis for reconstructing the legal framework.

The legal materials used in this research consist of primary, secondary, and tertiary legal materials. Primary legal materials comprise statutory and regulatory instruments, whereas secondary legal materials consist of approximately 30 scholarly sources, including accredited national journals, reputable international journals, books, and other academic publications issued primarily between 2019 and 2026, selected based on their relevance to the research topic. Data were collected through documentary research, while the legal materials were analyzed using a descriptive-analytical qualitative method by identifying, classifying, interpreting, and comparing relevant legal norms and conceptual frameworks. The analysis subsequently develops prescriptive legal arguments to formulate a reconstructed legal framework governing post-mortem access to Electronic Medical Records through the integration of the post-mortem privacy concept and blockchain technology.

III. RESULTS OF RESEARCH AND DISCUSSION

3.1. The Legal Framework Governing Post-Mortem Access to Electronic Medical Records in Indonesia

The advancement of digital transformation in the healthcare sector has accelerated the transition from conventional medical record systems to Electronic Medical Records (EMRs) integrated within the national healthcare ecosystem. This digitalization has enhanced the effectiveness of healthcare services by facilitating access to medical information, ensuring continuity of care, and improving administrative efficiency. However, as EMRs contain highly sensitive health data, they also increase the risk of privacy violations if not accompanied by an adequate legal protection framework (Azmi, Rahayu, & Babo, 2024). Accordingly, Indonesian law recognizes EMRs not merely as administrative healthcare documents but also as information embodying patients' privacy rights that must be legally protected. This issue becomes even more complex upon a patient's death, as it requires balancing the obligation to maintain the confidentiality of medical records with the legitimate interests of family members, heirs, and other parties who are legally entitled to access the patient's health information.

The legal protection of Electronic Medical Records (EMRs) in Indonesia is normatively established through three principal legal instruments:

- a) Law Number 17 of 2023 on Health;
- b) Law Number 27 of 2022 on Personal Data Protection ,dan
- c) Minister of Health Regulation Number 24 of 2022 on Medical Records.

Collectively, these legal instruments establish a complementary legal framework governing the implementation of Electronic Medical Records, the obligation to maintain the confidentiality of health information, and the procedures for the use of medical data under specific circumstances. Accordingly, the protection of Electronic Medical Records extends beyond the scope of health law and also forms an integral part of Indonesia's personal data protection regime. This approach reflects the recognition of health data as an individual right that must be safeguarded against any processing or disclosure lacking a lawful legal basis (Lestari et al., 2024).

Law Number 17 of 2023 on Health provides the legal foundation for the implementation of Electronic Medical Records (EMRs) as an integral component of healthcare services. Under this law, healthcare facilities and healthcare professionals are required to create, maintain, manage, and preserve the confidentiality of patients' medical records in accordance with the applicable standards of healthcare services. The recognition of EMRs under this legislation demonstrates that electronic medical records possess the same legal status as conventional medical records, thereby ensuring that all information contained therein is afforded the same level of legal protection (Benseghir, Zerara, Bentría, Bendriss, & Muhtar, 2025). In addition to serving as medical documentation, EMRs also possess evidentiary value in the resolution of legal disputes and provide an essential basis for clinical decision-making. Therefore, the implementation of EMRs must be accompanied by legal safeguards that ensure the integrity, security, and confidentiality of patients' health data (Mandey, 2025b).

The protection of health information is further strengthened by Law Number 27 of 2022 on Personal Data Protection, which classifies health data as specific personal data (Mandalika, Banu,

Irzaqie, & Muaja, 2025). This classification reflects the higher degree of sensitivity associated with health data compared to general personal data, as it relates directly to an individual's physical and mental condition. Consequently, any processing of health data must be based on the data subject's consent or another lawful basis provided by legislation and must comply with the principles of legality, transparency, purpose limitation, security, and accountability. In the context of Electronic Medical Records (EMRs), the Personal Data Protection Law extends legal protection beyond the traditional obligation of medical confidentiality to encompass the protection of patients' privacy rights as data subjects. (Setiyanta & Dewi, 2025).

Minister of Health Regulation Number 24 of 2022 on Medical Records, as the implementing regulation, provides detailed rules governing the management of Electronic Medical Records (EMRs), including their creation, storage, data exchange, and the disclosure of medical record information (Saputra, 2024). This regulation affirms that medical records are confidential documents that must be protected, as they contain information relating to a patient's identity, medical history, examination results, diagnosis, medical treatment, and therapy. Accordingly, healthcare facilities are required to implement security systems capable of ensuring the confidentiality, integrity, and availability of data throughout the operation of Electronic Medical Records (EMRs). These provisions demonstrate that Indonesian law recognizes the confidentiality of medical records as a fundamental principle of digital healthcare services and as an essential safeguard for patients' privacy rights (Mandalika et al., 2025).

Although the principle of confidentiality constitutes the primary foundation for the management of Electronic Medical Records (EMRs), Indonesian law permits certain parties to access medical information under specific circumstances as provided by legislation (Indra, Dewi, & Wibowo, 2024). As a general rule, patients, as data subjects, have the primary right to access their own medical records. In addition to patients, access to Electronic Medical Records may also be granted, under certain conditions, to immediate family members or heirs, law enforcement authorities for the purposes of judicial proceedings, researchers who satisfy applicable ethical and data confidentiality requirements, and other parties authorized by law. Nevertheless, such access rights are not absolute, as they remain subject to the intended purpose of use, the principle of proportionality, and the continuing obligation to maintain the confidentiality of patients' health information (Mubarak, Wiradirja, & Pranadita, 2025).

The regulation governing access to Electronic Medical Records (EMRs) after a patient's death is only partially addressed under Minister of Health Regulation Number 24 of 2022 on Medical Records. Article 32 affirms that the obligation to maintain the confidentiality of the contents of medical records continues to apply even after the patient's death. This provision indicates that the duty of confidentiality between healthcare professionals and patients does not cease upon death but remains binding as part of professional obligations and the protection of the patient's dignity. On the other hand, Articles 34 to 36 permit the disclosure of medical record information under specific circumstances, including for the purposes of law enforcement, healthcare audits, research, education, and other purposes authorized by

law. Accordingly, confidentiality remains the general rule, while the disclosure of medical record information constitutes an exception that must strictly comply with the applicable legal requirements.

Based on the foregoing discussion, it is evident that the regulatory gap concerning post-mortem access to Electronic Medical Records (EMRs) extends beyond the absence of legal recognition of the concept of post-mortem privacy. It also reflects the lack of a legal mechanism that enables patients to determine who may access their health data after death as an exercise of patient autonomy. Unlike previous studies, which have primarily focused on medical record confidentiality, personal data protection, or the security of health information systems, this study proposes a reconstructed legal framework that places patients' wishes at the center of post-mortem access to Electronic Medical Records. This reconstruction is proposed through a blockchain-based Digital Health Advance Directive mechanism, which enables patients' consent to be permanently recorded (immutable), verified through an audit trail, and automatically implemented in accordance with the authorization determined in advance. Accordingly, blockchain technology is not presented as the ultimate objective of this study but rather as a supporting instrument for promoting legal certainty, respecting patient autonomy, and protecting post-mortem privacy in the governance of Electronic Medical Records.

3.2. Patient Autonomy in Determining Access to Post-Mortem Electronic Medical Records: A Post-Mortem Privacy Perspective

Patient autonomy is one of the fundamental principles of bioethics and health law, recognizing patients as legal subjects who have the right to self-determination (Lewis & Holm, 2022). According to Beauchamp and Childress, respect for autonomy requires that decisions concerning an individual be made freely, voluntarily, and with full understanding, free from coercion (Childress & Beauchamp, 2022). In healthcare practice, this principle is implemented through the doctrine of informed consent, which recognizes patients' right to accept, refuse, or choose the medical treatment to which they will be subjected. However, developments in modern health law demonstrate that the scope of patient autonomy extends beyond consent to medical treatment alone. It also encompasses patients' right to control their personal health information, including the authority to determine who may access, use, or obtain their Electronic Medical Records (EMRs) containing sensitive personal data (Wilkinson, Levy, & Allen, 2025).

This understanding demonstrates that Electronic Medical Records (EMRs) serve not only administrative and clinical functions but also embody patients' privacy rights. Information concerning an individual's medical history, diagnosis, treatment, psychological condition, and genetic data constitutes part of their personal identity, the protection of which is inherently linked to human dignity (Kovács, 2024). Therefore, control over personal health information forms an essential aspect of the exercise of patient autonomy. The article *Impact of Nurse-Patient Relationship on Quality of Care and Patient Autonomy in Decision-Making* emphasizes that healthcare relationships should not be

dominated by healthcare professionals but should instead provide patients with the opportunity to make decisions independently. In this context, respect for patient autonomy extends beyond offering choices regarding medical treatment; it also requires recognition of patients' right to determine how their health information is used and to whom such information may be disclosed (Molina-Mula & Gallo-Estrada, 2020).

Nevertheless, a legal issue arises once the patient has died. Traditionally, death has been regarded as terminating an individual's status as a legal subject, with the consequence that personal rights are generally considered to cease as well. This perspective raises a fundamental question: does a patient's right to determine the use of their health information also automatically end upon death? The rapid development of digital technology suggests that the answer to this question is no longer as straightforward as under the conventional legal approach. Digital data, including Electronic Medical Records (EMRs), remain stored, may continue to be accessed, copied, and even used long after the data subject's death. This situation has given rise to an ongoing debate as to whether an individual's interests in their personal data should continue to receive legal protection after death.

This debate has led to the development of the concept of post-mortem privacy, which is based on the premise that an individual's personal information continues to warrant legal protection even after death. Such protection is intended not only to preserve the confidentiality of personal data but also to respect the individual's dignity, reputation, wishes expressed during their lifetime, and the legal interests that may continue to exist after death (Harbinja, McVey, & Edwards, 2024). Within the digital society, this concept has evolved alongside discussions on digital legacy, encompassing the management of digital accounts, electronic mail, social media, cloud storage, and health data that continue to exist after the death of their owner. Research on post-mortem data demonstrates that the central issue is no longer whether such data continue to exist, but rather who has the legitimate authority to determine their access, use, and management.

The concept of post-mortem privacy is closely linked to the principle of patient autonomy. If patients are recognized during their lifetime as having the right to make decisions regarding medical treatment, choose their healthcare professionals, give or withhold consent, and control their personal health information, then, from a philosophical perspective, there is no compelling justification for disregarding their wishes concerning the use of Electronic Medical Records (EMRs) after death (Arthanti, 2025). Accordingly, the legal issue is no longer merely whether Electronic Medical Records should remain confidential following a patient's death, but rather who has the authority to determine which parties may access them. This shift in perspective is significant because it distinguishes between the right of access to Electronic Medical Records and the right to determine access. The latter constitutes a direct manifestation of patient autonomy, yet it has not been adequately recognized or regulated within the Indonesian legal system (Bak & Willems, 2022).

An analysis of the Indonesian legal framework shows that Law Number 17 of 2023 on Health, Law Number 27 of 2022 on Personal Data Protection, and Minister of Health Regulation Number 24 of 2022 on Medical Records provide legal protection for the confidentiality of Electronic Medical Records (EMRs). In particular, Article 32 of Minister of Health Regulation Number 24 of 2022 expressly provides that the confidentiality of medical records must be maintained even after a patient's death. Nevertheless, these legal instruments do not establish a mechanism that enables patients to determine during their lifetime who may or may not access their Electronic Medical Records after death. The provisions governing the disclosure of medical records primarily rely on the consent of the patient, immediate family members, or heirs under certain circumstances, without explicitly regulating the legal effect of the patient's wishes where they conflict with the wishes of other parties. Accordingly, the Indonesian legal framework continues to place greater emphasis on the principle of confidentiality than on the principle of patient autonomy.

This approach may be compared with the legal framework in Malaysia. Although the Personal Data Protection Act 2010 and its subsequent developments do not recognize deceased individuals as data subjects, the Malaysian legal system nevertheless recognizes the legal significance of consent given during an individual's lifetime in certain contexts. One example is the Human Tissues Act 1974, which regulates the use of human tissue after death based on the individual's prior consent or the authority of designated persons. This demonstrates that, although Malaysia has not expressly recognized post-mortem privacy as an independent privacy right, it nevertheless acknowledges that an individual's wishes expressed before death may continue to have legal relevance in specific circumstances. This experience suggests that the protection of post-mortem interests does not necessarily require an expansion of the legal definition of privacy rights but may instead be achieved through legal mechanisms that respect the preferences or consent expressed during an individual's lifetime.

Based on the foregoing discussion, it can be concluded that patient autonomy in determining access to Electronic Medical Records (EMRs) after death has not yet received adequate legal protection under the Indonesian legal system. While the existing legal framework ensures the confidentiality of health data following a patient's death, it does not provide patients with the opportunity to explicitly determine the future use of their health information after death. As a result, a regulatory gap exists regarding the relationship between the patient's wishes, the authority of family members or heirs, and the access rights of other parties to Electronic Medical Records. Therefore, a reconstruction of the legal framework is required not only to preserve the principle of confidentiality but also to recognize patient autonomy through the post-mortem privacy approach. Such a reconstruction provides the conceptual foundation for the use of blockchain technology as a mechanism capable of securely, transparently, and accountably recording, verifying, and implementing patients' wishes after their death.

3.3. A Model for Reconstructing the Legal Framework Governing Post-Mortem Access to Electronic Medical Records to Accommodate Patient Autonomy through the Post-Mortem Privacy Approach and Blockchain Technology

Based on the analysis of the existing legal framework and the concept of patient autonomy presented in the previous section, it can be concluded that the Indonesian legal system has established legal protection for the confidentiality of Electronic Medical Records (EMRs) through Law Number 17 of 2023 on Health, Law Number 27 of 2022 on Personal Data Protection, and Minister of Health Regulation Number 24 of 2022 on Medical Records (Amin, Setyonugroho, & Hidayah, 2021). Nevertheless, this legal protection remains primarily focused on the principle of confidentiality and has yet to accommodate the dimension of patient autonomy in determining the future use of patients' health data after death.

The current legal framework does not provide a legal mechanism that enables patients to expressly designate the parties who may or may not access their Electronic Medical Records (EMRs) after death. Consequently, uncertainty arises regarding the legal effect of patients' wishes where they conflict with the interests of family members, heirs, or other parties who are granted access under applicable legislation. This condition demonstrates that the existing legal framework has not yet fully incorporated the concept of post-mortem privacy, which requires respect for an individual's interests and preferences concerning their personal information after death. Therefore, a reconstruction of the legal framework is necessary not only to preserve the principle of confidentiality of Electronic Medical Records but also to integrate the principle of patient autonomy as the foundation for determining post-mortem access. Such a reconstruction is expected to achieve a balanced framework that safeguards patients' rights, provides legal certainty for healthcare facilities, and enhances accountability in the governance of Electronic Medical Records through the support of appropriate technological mechanisms.

The reconstruction of the legal framework governing post-mortem access to Electronic Medical Records (EMRs) cannot be achieved merely by introducing provisions identifying the parties entitled to access patients' health data. Rather, such a reconstruction must be founded upon legal principles capable of addressing the shortcomings of the existing regulatory framework while accommodating developments in information technology within healthcare services (Sudra, 2021). Accordingly, the proposed legal reform should not be directed solely toward filling existing regulatory gaps but should also ensure that the resulting framework safeguards patients' rights, provides legal certainty for healthcare facilities, and promotes accountability in the governance of health data. Based on the foregoing analysis, at least five fundamental principles should serve as the foundation for the proposed regulatory model: respect for patient autonomy, post-mortem privacy, legal certainty, accountability, and data security.

The first principle is respect for patient autonomy. This principle requires that patients be recognized not merely as objects of legal protection but as legal subjects with the authority to determine the future use of their health information, including after death. Under the current legal framework, greater emphasis is placed on the authority of family members or heirs to consent to the disclosure of Electronic Medical Records (EMRs), whereas the wishes that patients may have expressed during their lifetime have not been expressly recognized (Narendra, Dewi, & Ismiyati, 2020). As a result, decision-making authority regarding access to Electronic Medical Records may shift from the patient to other parties. Therefore, the proposed legal reconstruction should establish a mechanism that enables patients, during their lifetime, to directly designate the parties who may or may not access their Electronic Medical Records as a concrete manifestation of the principle of self-determination.

The second principle is post-mortem privacy. Unlike the traditional concept of confidentiality, which primarily focuses on prohibiting the unauthorized disclosure of information, post-mortem privacy recognizes the protection of patients' interests after death as an essential aspect of respecting human dignity and individual preferences. In the context of Electronic Medical Records (EMRs), such protection is intended not only to prevent the misuse of health data but also to ensure that patients' wishes regarding the use of their medical information continue to be respected after death (Hurruzia & Adisasmito, 2025). Accordingly, the proposed legal reconstruction should not merely preserve the statutory obligation to maintain the confidentiality of Electronic Medical Records (EMRs), but should also recognize that patients retain a legitimate legal interest in determining who may access their health information after death.

The third principle is legal certainty. A regulatory framework that fails to establish a clear mechanism governing the relationship between patients' wishes, the rights of family members, the authority of heirs, and the interests of other parties may create legal uncertainty in healthcare practice. Healthcare facilities are often confronted with the dilemma of either complying with requests from family members or preserving the confidentiality of Electronic Medical Records (EMRs), as no legal provision expressly establishes the priority among these competing interests (Darmawan, 2023). Therefore, the proposed legal reconstruction should establish a mechanism that provides clear guidance regarding the parties entitled to access Electronic Medical Records based on the patient's previously expressed preferences. Such legal certainty not only safeguards patients' rights but also provides healthcare facilities with a clear legal basis for making decisions.

The fourth principle is accountability. Every act involving the disclosure, use, or denial of access to Electronic Medical Records (EMRs) must be legally accountable through a transparent and auditable recording mechanism. Accountability is particularly important because health data constitute a category of specific personal data and are therefore highly sensitive. Without an adequate recording mechanism, unauthorized access or data alteration may be difficult to establish in the event of a legal

dispute. Accordingly, the Electronic Medical Records management system should be capable of documenting all access activities so that every action is supported by a verifiable digital audit trail.

The final principle is data security. The protection of Electronic Medical Records (EMRs) is not determined solely by the existence of legal norms but also by the capability of technological systems to preserve the integrity, confidentiality, and authenticity of data. The risks of data breaches, manipulation, and unauthorized alteration of health information demonstrate that legal protection must be accompanied by the strengthening of technological security mechanisms (Dady, 2026). In this context, blockchain technology is not positioned as the ultimate objective of legal reconstruction but rather as an instrument capable of supporting the implementation of these five principles. The characteristics of blockchain, including immutable ledger technology, cryptographic mechanisms, permissioned access, smart contracts, and audit trails, enable patients' wishes expressed during their lifetime to be authentically recorded, protected against unauthorized modification, and verified whenever access to Electronic Medical Records is requested after death. Accordingly, the adoption of blockchain technology gains legal legitimacy not merely from technological advancement itself, but from its capacity to address legal needs that cannot yet be adequately resolved by the existing regulatory framework (Syukurillah, Lutviana, & Purwono, 2024).

Based on the weaknesses of the existing legal framework, this study proposes a reconstruction of the regulatory mechanism governing post-mortem access to Electronic Medical Records (EMRs) that prioritizes respect for patient autonomy. According to the author, such a mechanism should begin when patients register with healthcare facilities by providing them with the opportunity to express their wishes regarding the management of their Electronic Medical Records after death. These wishes should be documented in the form of an electronic advance directive, which constitutes an integral part of the Electronic Medical Records system. Through this mechanism, patients may consciously determine the parties who are granted access rights as well as those who are excluded from accessing their health information (Tampubolon, Bhawiyuga, & Siregar, 2022). Accordingly, access to Electronic Medical Records after a patient's death would no longer be based solely on the consent of family members or heirs but would first be guided by the patient's legally expressed wishes during their lifetime.

According to the author, this model is capable of shifting the regulatory orientation from merely protecting confidentiality toward respecting patient autonomy as the primary holder of interests in personal health information. To ensure legal certainty and effective implementation, patients' electronic expressions of intent should be stored through a permissioned blockchain-based system. The utilization of blockchain is not intended to replace the existing Electronic Medical Records (EMRs) system, but rather to function as a governance layer that permanently records patients' consent in an encrypted, tamper-resistant (immutable) manner and enables verification through an audit trail (Sonia, Putra, & Nurmalasari, 2026). Accordingly, every request for post-mortem access to Electronic Medical Records can be verified based on the patient's documented wishes, thereby establishing a balance between the

protection of patients' rights, legal certainty for healthcare facilities, and accountability in health data governance.

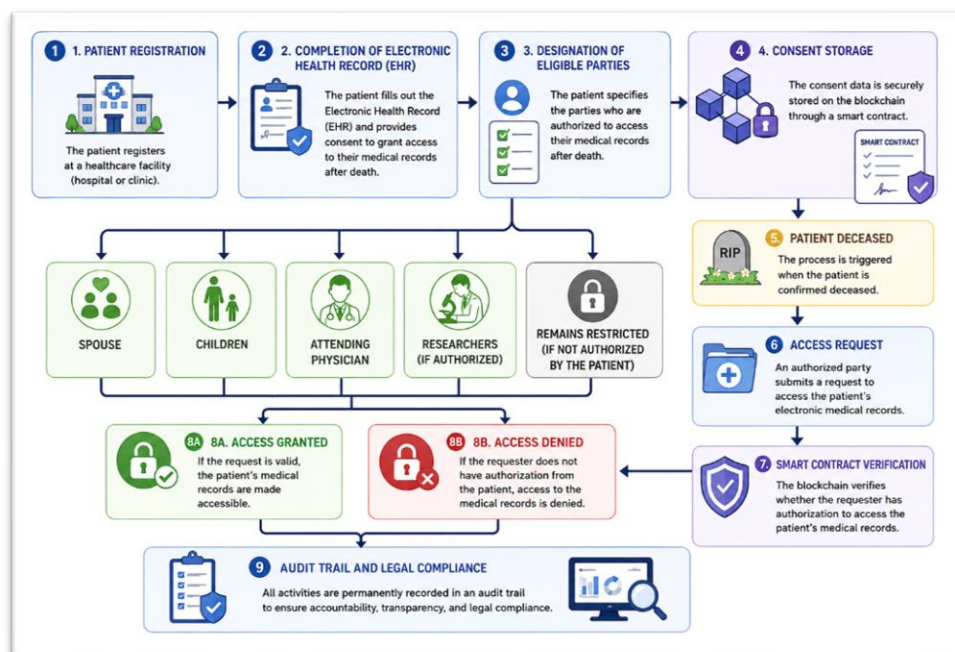


Figure 1. Conceptual Model for the Reconstruction of Post-Mortem Access Regulation for Electronic Medical Records

The utilization of blockchain technology in the proposed model for reconstructing the regulation of post-mortem access to Electronic Medical Records (EMRs) is not based solely on its status as a technological innovation, but rather on its capability to address legal needs that cannot yet be fulfilled by the current Electronic Medical Records management system. According to the author, the primary issue does not concern the storage of health data itself, but rather the mechanism for proving and implementing patients' wishes after death. Therefore, blockchain technology is positioned as a supporting instrument that ensures patients' electronic advance directives are stored authentically, protected from unilateral alteration through an immutable ledger, and remain verifiable when requests for access to Electronic Medical Records arise. These characteristics provide assurance that patients' preferences expressed during their lifetime remain the primary basis for decision-making, thereby ensuring that respect for patient autonomy does not cease upon death but continues to be protected through a technological mechanism with strong evidentiary value and data integrity.

In addition to maintaining data integrity, blockchain technology also supports the realization of the principles of legal certainty, accountability, and security in the management of Electronic Medical Records (EMRs). Through the smart contract mechanism, the system can automatically verify each access request based on the preferences previously established by the patient, thereby reducing the possibility of inconsistent interpretations by healthcare facility administrators (Ramadahni, 2024). Furthermore, the audit trail feature enables every access activity, modification of authorization, and denial of requests to be permanently documented and subject to audit in the event of a legal dispute.

The use of cryptographic mechanisms and permissioned blockchain systems also provides protection against the risks of data manipulation and unauthorized access, ensuring that only parties authorized by the patient's wishes or by applicable legal provisions may access such information t(Setiawan, Bhawiyuga, & Siregar, 2020). Accordingly, the author argues that blockchain does not replace the function of law as the foundation for granting access rights; rather, it strengthens the implementation of legal norms through a system that is more transparent, secure, accountable, and capable of ensuring the consistent realization of patients' wishes.

Table 1. The Role of Blockchain in Legal Reconstruction

Blockchain Feature	Technical Function	Supported Legal Value
Immutable ledger	Data cannot be altered	Legal certainty and evidentiary integrity
Smart contract	Automatic execution based on predefined requirements	Respect for patient autonomy
Audit trail	All activities are permanently recorded	Accountability and evidentiary support
Cryptography	Encryption and access control mechanisms	Confidentiality and personal data protection
Permissioned Blockchain	Restricted access limited to authorized parties	Protection of patient rights and data security

Based on the analysis of the regulatory gaps concerning patient autonomy in determining access to Electronic Medical Records (EMRs) after death, the author argues that legal reconstruction must be conducted comprehensively and should not be limited merely to amendments within a single statutory regulation. Such reconstruction should encompass the reformulation of legal substance, the strengthening of healthcare governance, and the integration of technological mechanisms capable of ensuring the effective implementation of patients' wishes. Therefore, several aspects require reconstruction in order to optimally realize the protection of patients' rights, legal certainty, and accountability in Electronic Medical Records management, as follows:

- a) Reconstruction of Norms under Minister of Health Regulation Number 24 of 2022
 Minister of Health Regulation Number 24 of 2022 has regulated the confidentiality and disclosure of Electronic Medical Records; however, it does not explicitly recognize the patient's right to determine the parties who are entitled or not entitled to access Electronic Medical Records after death. According to the author, additional norms are required to accommodate electronic advance directives as the primary legal basis for granting post-mortem access to Electronic Medical Records.

b) Strengthening the Principle of Patient Autonomy in the Health Law

In addition to amendments at the regulatory level, legal reconstruction must also be supported by strengthening the norms contained in Law Number 17 of 2023 on Health. Such regulation should affirm that respect for patient autonomy does not cease at the stage of providing consent for medical treatment, but also includes the patient's right to determine the use of health information after death as part of the protection of personal data rights and human dignity.

c) Development of Standard Operating Procedures in Healthcare Facilities

According to the author, regulatory changes will not be effective without clear operational mechanisms. Therefore, every healthcare facility should establish standard operating procedures governing the recording, modification, revocation, and implementation of patients' electronic advance directives. These procedures should also regulate the verification of the identity of parties requesting access, mechanisms for granting or denying access, and dispute resolution procedures in cases where patients' wishes conflict with the interests of family members or heirs.

d) Development of Blockchain-Based Consent Management Systems

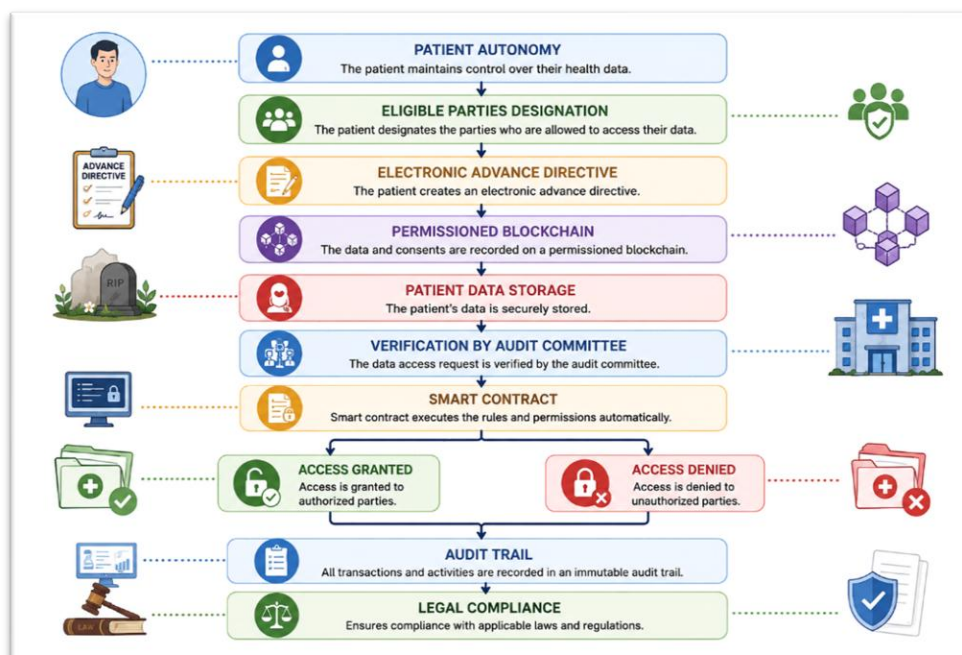
As an implementation of the proposed legal reconstruction, the Electronic Medical Records system should be developed by incorporating a permissioned blockchain-based consent management feature. This feature functions to permanently record patient consent, verify each access request based on predetermined preferences, and document all access activities through an audit trail mechanism. According to the author, blockchain technology is not intended to replace the existing Electronic Medical Records system but rather to serve as an instrument that strengthens the implementation of legal norms through a secure, transparent, and accountable system.

Based on the above explanation, the reconstruction of the legal framework governing post-mortem access to Electronic Medical Records cannot be achieved merely through the introduction of new legal norms but requires harmonization between regulations, institutional governance, and technological support. According to the author, the synergy of these three aspects will produce a system that is more adaptive to developments in information technology, respects patient autonomy, provides legal certainty for healthcare facilities, and strengthens the protection of post-mortem privacy in Indonesia.

As a form of reconstruction of the existing legal framework, the author proposes a conceptual model for managing post-mortem access to Electronic Medical Records based on the principle of patient autonomy, the concept of post-mortem privacy, and supported by blockchain technology as an implementation instrument. This model is developed based on the finding that the primary problem in Indonesia's Electronic Medical Records regulation does not lie in the obligation to maintain patient

confidentiality, but rather in the absence of a legal mechanism that grants patients authority to determine which parties may access their health information after death. Therefore, the starting point of reconstruction is not the implementation of blockchain technology, but the recognition of patients' rights to determine the future management of their Electronic Medical Records while they are still alive. Such recognition is realized through an electronic advance directive, which becomes an integral part of the Electronic Medical Records system and serves as the primary basis for granting or denying access to patients' health data after death.

According to the author, blockchain should not be positioned as the objective of legal reconstruction, but rather as an instrument that ensures the effective implementation of legal norms. After patients provide their electronic advance directives, these preferences are stored within a permissioned blockchain system, providing authenticity, encryption, resistance to unilateral alteration (immutable ledger), and verifiability whenever requests for access to Electronic Medical Records arise. Subsequently, after the patient's death, healthcare facilities verify access requests by referring to the preferences recorded on the blockchain. This process is then executed through a smart contract mechanism that automatically adjusts the granting or denial of access based on the patient's wishes and applicable legal provisions. All subsequent access activities are documented through an audit trail, ensuring legal accountability in the event of disputes. Therefore, the proposed conceptual model not only strengthens the protection of post-mortem privacy, but also establishes an Electronic Medical Records governance system that is more transparent, accountable, secure, and capable of providing legal certainty for patients, healthcare facilities, and other relevant stakeholders.



Gambar 2. Model Konseptual Rekonstruksi Pengaturan Akses Rekam Medis Elektronik Pasca-Kematian Berbasis Otonomi Pasien melalui Teknologi Blockchain

IV. CONCLUSION

Based on the discussion, it can be concluded that the legal framework governing post-mortem access to Electronic Medical Records (EMRs) in Indonesia has provided protection for the confidentiality of patients' health information through Law Number 17 of 2023 on Health, Law Number 27 of 2022 on Personal Data Protection, and Minister of Health Regulation Number 24 of 2022 on Medical Records. However, these regulations remain predominantly oriented toward the principle of confidentiality and have not fully accommodated the principle of patient autonomy, particularly regarding the authority of patients to determine which parties may or may not access their Electronic Medical Records after death. This condition demonstrates the existence of a legal gap concerning the status of patients' expressed wishes, the scope of family or heirs' authority, and the mechanism for granting access to post-mortem health data. From the perspective of post-mortem privacy, death does not automatically eliminate an individual's legal interests in personal information; therefore, patients' preferences expressed during their lifetime remain legally relevant and should be respected as part of the protection of human dignity and privacy rights.

Based on these findings, the reconstruction of the legal framework governing post-mortem access to Electronic Medical Records should not merely maintain the obligation to preserve data confidentiality but must also provide a mechanism that enables patients to determine the future management of their health information after death. Such reconstruction may be realized through the recognition of an electronic advance directive as the primary basis for granting or denying access to EMRs, strengthening patient autonomy within health regulations, establishing operational procedures in healthcare facilities, and developing a permissioned blockchain-based consent management system. In this model, blockchain technology is not positioned as a substitute for legal norms, but rather as a supporting instrument capable of ensuring the authenticity of patients' wishes, providing an immutable ledger, maintaining audit trails, and strengthening secure and accountable access control mechanisms. Therefore, the integration of legal regulation, institutional governance, and blockchain technology may establish a post-mortem Electronic Medical Records management system that balances the protection of patient rights, legal certainty for healthcare providers, and the realization of post-mortem privacy within Indonesia's digital healthcare ecosystem.

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